

**MANUAL
FOR
OPERATIONAL MAINTAINCE
OF
CONVEYOR BELTS
HINDUSTAN RUBBERS (SILVASSA)**

A. ACCEPTANCE OF BELT

The Belt is supplied in wooden crates or in spool. At the starting end of belt, there is an INDUS logo embossed on belt surface, which is required to be checked.

It should be ensured whether the belt is supplied in crate/spool in properly packed condition.

B. BELT STORAGE

Handling

Conveyor Belts are dispatched in wooden crates. All crates are marked with an arrow sign showing the direction in which they are to be rolled or hoisted. Belts should always be moved by lifting and transporting rather than rolling, due to the danger to men and possibility of damage to the belt, if collapsing as a result of being rolled. Needless to emphasize the belt edge should be protected carefully.

Storing

Unless damaged in transit, Belt should be kept in upright position until it is being commissioned. The belts are to be stored in a cool, dry, covered and well ventilated place away from direct sunlight. Other factors like exposure to sunlight, ozone in sea area, contact with all types of solvents and corrosive liquids should be avoided.

If proper warehousing is not possible, then the belt should be covered with black polythene, suitable fastened against wind damage. In case of belts weighing excess of 1.5 tons and not installed within 2 or 3 months and beyond, it should be supported off the ground on "A" frames as shown in Fig. A. Belts kept as spares should definitely stored off the ground and rotated 1/4 turn every six months. Whenever lengths are cut off from the belting roll, the end should be sealed with Conveyor Belting repair solution to prevent ingress of moisture into the carcass.

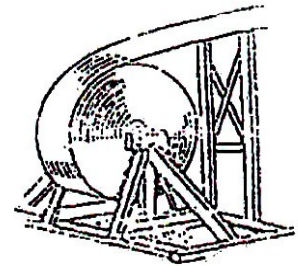


Figure-A

The Drums in storage should be examined once in 2 or 3 months to ensure that packing is not failing due to white ants and similar insects. Spraying of DDT over packing will prevent this type of damage.

C. BELT INSTALLATION

a) Pulley Lagging

Following reasons necessitate the use of pulley lagging for driving pulley;

- i) It facilitates the belt driven by lower slack side tension thus resulting in lower total tension in many a case because of improved co-efficient of friction.
- ii) Slippage due to wet condition for grooved lagging
- iii) Improvement for life for pulley.

b) Checks on idlers

Positioning the idlers normal to the feet of travel is a must.

Gaps (=10mm) between the idlers as per specification should be maintained to avoid nipping of belt.

The distance between the idlers on carrying side and that on return side should be well maintained.

The movement of idlers according to direction of belt movement will help in centering the belt, if tracking on one side is noticed.

A good deal of alignment effect is experienced upon lifting the toughing idlers which should be limited within 2°. Tilt greater than 2° may lead to undesired wear of pulley cover.

Return idlers are flat. It has no self alignment tendency whatsoever. However, any shift of their axis with respect to path of belt will bring about perpetual ameliorative effect in one direction,

Slight advancement and lift of alternate ends of return rolls nearest to tail pulley will bring forth centering attenuation of belt. Rotation about center vertical pivot of carrying idlers caused by off-centered running of belt, will bring the belt back to center. Grease used for lubricating the idlers should not contaminate the surface of conveyor for its inherent detrimental effect.

c) Unrolling

Unrolling of belt should be done with belting care so as to avoid any undesirable damage of belt during unwinding. The drum should be hoisted for placement of bar through the centre. The edges of the belt should be properly protected.

During drawing of belt for unrolling, care should be taken that undesired additional rotation will loosen the residual remains of belts in drum for tight batching. End of belt should be well protected from ingress of moisture or foreign materials when ever part of length is cut from the roll.

d) Terminal Arrangement

In order to augment cleaning of spillage and dribble cleared by tail scrapers, tail pulley should have 0.6 mtrs clearance underneath.

Incase where materials of one conveyor is transferred to the other, the tail should have 1.8 - 3.0 m overlap so that material cleared by scrapers of upper conveyor can be offloaded on to the bottom conveyor.

The distance between top surfaces of two conveyors running one over the other at the point of transshipment, space will be min 6.0m more than the dia of pulley.

In transition zone, the edge of the belt is subjected to higher tension which can be reduced if top of head pulley levels to half of trough depth when transition distance should be more than $1.25 W$ (where W is the belt width).

When top of the head pulley corresponds to horizontal roll of troughing stand (full trough depth), the TD should be more than $1.75 W$.

e) Training

A belt is required to run straight and centrally for which necessary adjustment and correct alignment of idlers, pulleys and parallelism of shafts and even loading conditions may be necessary.

Head pulley need not be crowned. Crowned take up pulley will take care of misalignment in take up carriage.

Terminal, snub, bend and take up pulleys should be at 90° to belt centre line. Idlers and pulleys should be free to rotate and devoid of dirt proper alignment of troughing and return idlers should be ensured when the idler axis to be initially adjusted at 90° to the conveyor centre line i.e. to the line of belt travel.

Procedures of unidirectional adjustment

a) Centering of belt can be done by turning the idlers in the direction of belt travel

b) Centering of belt can be achieved upon tilting the troughing idlers forwarded not over 2° . Greater angular tilt will cause undesired wear of pulley side of belt. Unidirectional adjustment is not effective for reversible belt. Any shift of axis of return idlers with respect to the path of the belt will provide corrective effect in one direction.

Advancement or lift of alternate ends of return rolls in close proximity to the tail pulley will help in centering the belt.

The advent of self-aligning idlers (troughing and return rollers) is an outstanding milestone to have self-adjusting effect of belt centering. It is more effective when raised slightly above the two adjustment idlers.

Use of side guide rollers in compelling the belt to run straight will not prove to be very effective because of excessive edge wear.

The extent of lubrication should not be allowed to be source of undesired deterioration of belt caused by lubricants.

f) Loading

The centre of gravity of material being dumped on belt surface should correspond to lowest point in the troughing idlers.

Proper arrangement of chute will mitigate the sidewise movement of belt upon loading provided the belt is properly centered at its loading point

What is considered to be of utter importance is the loading point, as it suffers a major abuse of abrasion, impact and non-coherent adjustment of material off-loading speed and belt speed.

Incase of presence of Fines, the chute should be greater than twice the size of largest lump. But in case of uniform, particularly the chute should be minimum 3.5 times the lump size.

The width of discharge chute should be less than $\frac{2}{3} W$, where " W " is the width of receiving belt.

No part of the steel structure of the chute and skirt should come closer than 25mm to the belt surface.

Skirt rubber length is usually $4W$ or $5W$. The resistance offered by skirt on belt surface can be checked.

The slope of chute should be limited within 35° for coal and rock.

The severity of impact due to loading can be reduced by placing rubber covered disc idlers or pneumatic idlers. It can be reduced further upon allowing the lines to drop on belt first upon cutting the 'V' slot at bottom of chute.

g) Cleaning

Accumulation of materials on return rolls and snub pulleys will effect belt running of centered and damage against structure. Return idlers should be placed sufficiently below the structure that dirt accumulation on idlers and misalignment may be visible. Rubber scrapers (12-25mm thick) are clamped against bars upon extending the rubber by 2T (where T is the thickness of scrapers) Counter-weighted against belt surface. Placement of 3-4 steel scrapers in succession is usual practice. These are placed diagonally and mounted against leaf spring to pressurise the belt to scrape sticky materials off the belt surface. Rotation of bristle or rubber vane brushes by 4-5 times the belt speed will ensure removal of dry materials off the belt surface.

Cleaning is effective when performed between hand pulleys and snub pulleys. But sticky material needs scraping on head pulley itself to be dumped on chute. Accumulation of material on snub pulley sometimes can be avoided by soft rubber lagging or by scraping directly against pulley.

Build-up material and training problem can be avoided by rubber disc or spiral type return rolls.

Rubber faced plows held by gravity and placed (at an angle of 45°) immediately in front of tail pulleys are used to remove lumps which is going to be entrapped between belt and tail pulley.

h) Repairing

Repairing of damaged portion of the belt is given top priority for its inherent importance to avoid undesired propagation.

- i) With the help of rubber cutting knife, the rubber is cut bevelled of the carcass of the belt without damaging the carcass whatsoever.

Repair should be circular or oval in shape.

- ii) The cover rubber is to be removed by pincers upon dividing the strips by knife longitudinally
- iii) Solvent may be used to clean the exposed portion. The exposed canvas surface and bevelled cut edges may be roughened by wire brush.
- iv) The carcass must be devoid of moisture. 2-3 coats of rubber solution is to be applied. Each of the coat should be allowed to dry completely.
- v) Use cover rubber thickness of following thickness for repairing:

Finished Belt Cover Rubber Thickness (mm)	Input (Repair) Rubber Thickness (mm)
Upto 1.5	+0-1
>1.5	+0.2

- vi) Repair rubber is to be cut to correct shape and bevelled at edges and then painted with rubber solution.
- vii) The solutioned surface of repair rubber is to be placed against carcass and the portion is to be rolled with stitch wheel and pricked properly and the excess rubber or overloaded portion is to be trimmed off with knife.
- viii) Cure the portion in Press as detailed below:

Cover Repair	Pressure	Time	Temp.
</=6mm	10-12kg/cm ²	30min	140-145°
>6mm	10-12kg/cm ²	40min	140-145°
Cool it down to below 80°C prior to opening			

i) Fastner Joints

Steel Hinge

- i) "Alligator Lacing" is applied by Hammer-Driving the prongs into the body of the belt.

Either side of the belt is hinged with pin placed through alligator. Alligators are applied in small sections.

It resists troughing of narrow belt; it is used for low tensioned conveyor.

- ii) Hinged plate type U shaped clips are available in different sizes. End of the belt is bolted with clips. Holes are properly spaced by template and punched and belt ends are held by hinge pins placed through U shape clips.

It is used for medium tensioned conveyors.

Plate type

- i) 'Crescent' type consists of one or more plates slightly curved conforming pulley curvature. Plates are riveted across the belt joint.

These are placed on top side of the belt only. The rivets are hammered down to the other side.

It is used for transmission belt but not for reverse bends.

- ii) 'Flexco' and 'Plategrip' consists of matching pairs of bolted plates.

The plates are placed on either side of belt. Use of template for proper spacing of hole is a standard practice. The bolts are tightened enough to compress the belt between the plates deriving the holding power. It is accounted for leak proof joint having best abrasion and tension. It can be anti-sparking, anti-magnetic and corrosion resistant.

Use of smaller plates without sacrificing fastener holding ability is often preferred for its inherent advantage to avoid belt breakage. It is used for conveyor belting.

- iii) 'Steel pronged hook' type plate :

The plate side is placed on top side of belt and the prongs are driven down through the belt by hammer on pulley side of belt. It is used for conveyor belt of moderate tension but it is not as strong as the bolted plate type.

**ADVANTAGES
OF
CURING
BY
COLD VULCANISING
METHOD**

INTRODUCTION

The primary advantage of a cold splicing as opposed to hot splicing is saving in time and labour both in the transport and assembly of the vulcanizing press in curing time. Another advantage is the more favourable application at sites that are not readily accessible or on premises with a risk of fire-damp or explosion.

1. Selection of Technical Parameters.

- 1.1 Carefully go through the technical information regarding cold splicing of textile belts as given in Appendix A.
- 1.2 Procure the splicing materials and splicing tools.

2. Selection of Vulcanising Site.

- 2.1 Ensure proper means of access to the belt.
- 2.2 Ensure that the power supply line is adequate for lighting arrangement and other machine tools.
- 2.3 Ensure that the working area is properly ventilated.

3. Precautions to be taken.

- 3.1 Do not work in a confined unventilated area.
- 3.2 Avoid breathing the vapours of the solvents or cleaning agents.
- 3.3 Avoid contact of cement with skin and eyes.

4. Making the splice.

- 4.1 Tension the belt and arrest the two belt ends in front of the working area so that the tensioned belt cannot run back.
- 4.2 Align both ends of the belt so that they lie centrally on the working platform/ bottom platen of press over a length of 2 meters with one end overlapping the other.
- 4.3 Mark the appropriate areas which are to form the splice and draw centre line of each belt end extending about 2-3 meters on each end beyond the overlap portion.
- 4.4 Mark one of the belt ends at right angles with respect to the centre line.
- 4.5 Mark the bias length ($L=0.3 \times \text{Belt Width}$), on one belt edge and mark the bias angle with a chalked string.
- 4.6 Cut off the belt end on the bias with a scribing knife, (Fig. 1)
- 4.7 Calculate the additional length L and mark the distance L from belt end at both edges of the belt on top cover.
- 4.8 Join these two points. Draw a line parallel to this line at 25 mm interval on the top cover.
- 4.9 Pull off the 25 mm cover strip. (Fig-2)
- 4.10 Also mark the step length at both the edges and join them.
- 4.11 Lift one corner of the first fabric ply alongwith top cover by means of a ply cutter taking care not to cut more than one fabric type.
- 4.12 Cut out the cover and the first fabric ply into longitudinal strips with cobbler's knife. Remove the cover and the ply by pulling with a pincer. (Fig 3.)
- 4.13 Mark the next fabric step, cut into longitudinal strips and then strip it off.
- 4.14 Repeat these operations for each fabric step.
- 4.15 Level a 50 mm wide cover strip in front of the last fabric step (Fig. 4)
- 4.16 Remove any residual rubber from the exposed plies with a grinding wheel or a rotary wire brush taking care to prevent the wrap threads being damaged.
- 4.17 Clean the buffing dust in dry state with a soft brush.
- 4.18 Prepare the other half of the splice correspondingly.

5. Preparation of the Splicing Cement.

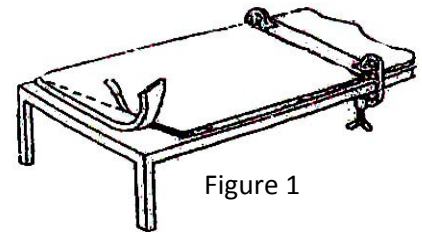


Figure 1

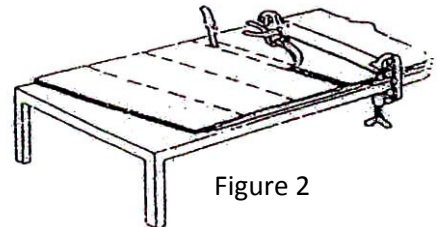


Figure 2

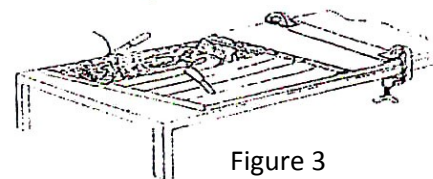


Figure 3

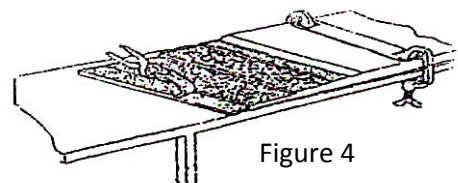
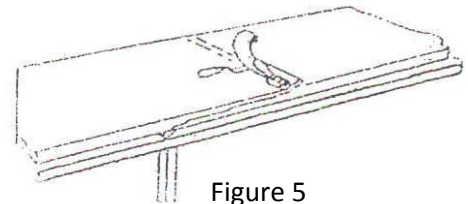


Figure 4

- 5.1 Pay strict attention to cleanliness.
- 5.2 Do not open the cement tin or hardener bottle until immediately before use. Since these have a very limited storage life in opened condition.
- 5.3 Do not mix the cold splicing cement with the hardener until immediately before use.
- 5.4 Just before use, pour 1 litre of cement into a clean container. Mix 1 bottle of hardener with it and stir well. (When 1 litre of container is procured, one bottle of hardener may be poured directly into it).
- 5.5 Store the balance quantity (4 litre) of inactivated cement for further use, after replacing the lid. Store it in a cool dry place.
- 5.6 Use the prepared cement within 2 hours.
6. Coating and Jointing the splice.
 - 6.1 Lay down both the ends of the spliced belt on each other to check for dimensional accuracy.
 - 6.2 Clean the two prepared surfaces once again with a soft brush.
 - 6.3 Coat the entire splice area, fabric steps, cut edges of the solid rubber and the 50 mm cover strip with the cold splicing cement by means of a soft bristled brush.
 - 6.4 Take care to spread out the coat well to prevent the formation of any globules of cement that fail to dry out of completely.
 - 6.5 Leave the coat to dry out thoroughly before the next coat is applied.
 - 6.6 Apply two coats for rubber surface and for finely structured fabric, but at least three coats for coarse fabric.
 - 6.7 Apply the last coat of cement simultaneously on both the halves to ensure that the two halves of the splice are ready for bonding at the same time.
 - 6.8 Ensure that the last coat is not completely dried out before jointing.
 - 6.9 Joint the two halves of the splice with meticulous care so that the single fabric steps butt precisely and at the same time stitch them down with a hand roller.
 - 6.10 Firmly stitch down the 50 mm cover strip on the 25mm bias channel and cut out the projecting cover strip with an angled knife. (Fig. 5).
 - 6.11 Leave the splice untensioned for about 4-6 hours before putting it back into operation.



APPENDIX A

TECHNICAL INFORMATION FOR COLD SPLICING

1. Splice Length.

The extra length of belting required to form the splice can be calculated from the following formula ;

$$\text{Additional Length (L1)} = 0.3 W + S.L (N-1) + 75 \text{ mm}$$

where, W = Belt width in mm

N = The number of carcass plies.

S.L. = Step Length in mm.

The step lengths for different belt constructions are given in Table 1.

2. Material Requirement

Grade of Belt Quantity

M-24, N17 5 litres &

HR 1 litre

FR 5 litres & 1 litre

❖ Generally 4.0Kg. of cold splicing cement is required for 1.0 M2, of splice area.

❖ 1 bottle of hardner per litre of solution supplied alongwith the kit.

3. Splicing tools required.

3.01	Knife with longer blade	3.15	Hand brush
3.02	Saddler's knife	3.16	3 meter measuring tape
3.03	Cobbler's knife with retractable blade	3.17	Trisquare
3.04	Ply cutting knife	3.18	Rasp File
3.05	Knife with flexible blade	3.19	Surface grinder
3.06	Angled knife	3.20	Soft-bristled brush
3.07	Quarter moon knife	3.21	Steel ruler
3.08	Whet stone	3.22	White chalked string
3.09	Knurled roller	3.23	Awl/pricker
3.10	Flat Roller	3.24	Iron mallet
3.11	Wire brush	3.25	Air bowler
3.12	Scissors	3.26	1 pair safety glasses
3.13	Pincers	3.27	1 pair safety gloves
3.14	Screw drivers		

TABLE 1

Description: Rubber Belt Spile Step Length

Construction: Listed below are the step length to be used when making a splice

FABRIC TYPE	RECOMMENDED STEP LENGTHS			
	2 PLY	3 PLY	4 PLY	5 PLY & MORE
PP 100 EP 100 28 Oz CN70	---	150	---	---
PP 125 PP 160 EP 125 EP 160 32 Oz 34 Oz CN 80	---	200	200	200
PP 200 EP 200	---	250	250	250
PP 250 EP 250 36 OZ 42 OZ 48 OZ CN 90 CN 105 CN 130 CN 175 CN 210	---	300	300	300
PP 315 EP 315	---	350	350	350
PP 350 EP 350 EP 400	---	400	400	400
EP 500	---	500	500	500

**ADVANTAGES
OF
CURING
BY
HOT VULCANISING
METHOD**

INTRODUCTION

A vulcanised splice is the strongest and most reliable method of jointing a belt and should be used whenever possible. It is nearly as strong as the belt itself, and is less troublesome than other fastening methods and results in longer belt life. The carcass of the belt is fully protected; the surface is left smooth, so that scrappers and brushes can be used more effectively and with less risk of damage to the belt than with a fastener.

1. Selection of Technical Parameter.

- 1.1 Carefully go through the technical information regarding splicing of belts as given in Appendix B.
- 1.2 Calculate splice length, procure the splicing materials and splice tools.
- 1.3 Determine the curing conditions e.g. time, temperature and pressure etc.

2. Selection of the Vulcanising Site

- 2.1 Ensure proper means of access to the belt.
- 2.2 Ensure that the power supply line is adequate for lighting arrangements and also for the vulcanizing press and other machine tools.

3. Precautionary Measures:

- 3.1 Do not work in a confined unventilated area.
- 3.2 Avoid breathing the vapours of the solvents or cleaning agents.
- 3.3 Avoid contact of cement with the skin and eyes.

4. Making the Splice:

- 4.1 Place the take up pulley in correct position. Tension the belt and arrest the two belt ends in front of the working area so that the tensioned belt cannot run back.

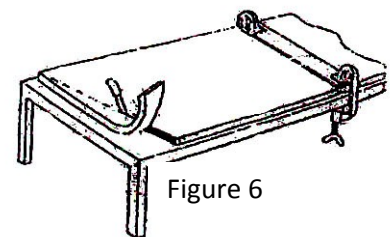


Figure 6

- 4.2 Align both ends of the belt so that they lie centrally on the working platform/bottom plates of press over-a-length of 2 meters with one end overlapping the other

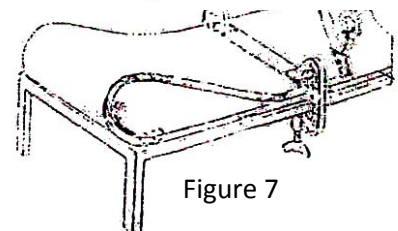


Figure 7

- 4.3 Mark the approximate areas which are to form the splice and draw centre line on each belt end extending about 2-3 meters on each and beyond the overlap portion.

- 4.4 Mark one of the belt ends at right angles with respect to the centre line

- 4.5 Mark the bias length ($0.3 \times \text{Belt Width}$), on one belt edge and mark the bias angle with a chalked string.

- 4.6 Cut off the belt end of the bias with a scribing knife.

- 4.7 Fold back the belt and draw a line parallel to the cut edge at a 25 mm distance on the bottom cover with the chalked string. (Fig. 7)

- 4.8 Pull off the 25 mm wide cover strip.

- 4.9 Buff the cut edge of the cover gently with a Wire brush or roughener.

- 4.10 Fold back the belt again and calculate the splice length L_s and mark at the both the edges of the belt

- 4.11 Join this line on top cover. Draw a line parallel to this at 25 mm distance on the top cover.

- 4.12 Pull off the line 25 mm cover strip. (Fig. 8)

- 4.13 Also mark the step length (SL) at both the edges and join them.

- 4.14 Lift one corner of the first fabric ply alongwith the top cover by means of a ply cutter taking care to cut more than one fabric ply.

- 4.15 Cut out the cover and the first fabric ply in longitudinal strips with a Cobbler's knife. Remove the cover and the ply by pulling with a pincer. (Fig- 9) taking care not to cut the ply below.

- 4.16 Mark the next fabric step cut into longitudinal strips and then strip it off.

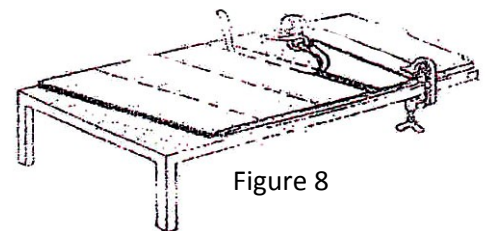


Figure 8

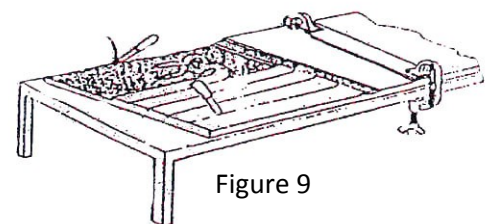


Figure 9

4.17 Repeat these operations for each fabric step.

4.18 Remove any residual rubber from the exposed area with a grinding wheel or a rotary wire brush taking care to prevent the warp threads being damaged. Clean buffing dust in dry state with a soft brush.

4.19 Prepare the other end of the splice correspondingly.

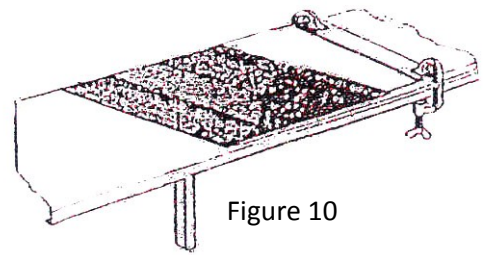


Figure 10

5. Preparation of the Splice

5.1 Lay down both ends of the spliced belt on each other to check for dimensional accuracy.

5.2 Clean the two prepared surfaces once again with a soft brush and apply two successive coats of splicing cement, the second coat being applied after the first one becomes completely dry.

5.3 Apply 0.5 mm skim gum to one of the coated belt from the cut edge of the top fabric ply onwards (Fig. 11) after the 2nd coat has thoroughly dried up.

5.4 Stitch well and remove the protective foil.

5.5 Build up the edges with two layers of approximately 10 mm wide skim gum.

5.6 Match up the two prepared surfaces again taking care that the belt ends match precisely and the single fabric steps butt precisely.

5.7 Stitch down firmly with a hand roller, working outwards from the belt centre to the edges. (Fig. 12)

5.8 Apply solution of the cover channels. Fill in the 50 mm wide (face and back) cover channel with cover rubber stock. (Fig. 13) cements and observes drying. Ensure that extra Rubber (about 1mm) is kept in width and thickness.

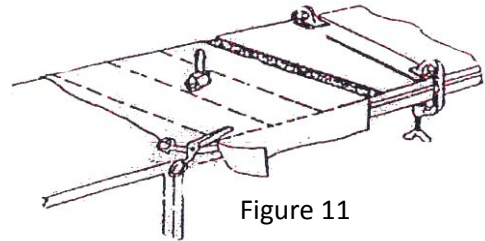


Figure 11

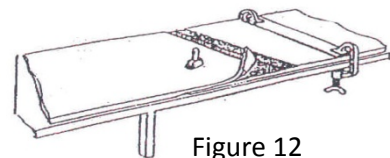


Figure 12

6. Vulcanization of the Splice.

6.1 Check the alignment of bottom traverse and heating platens (which constituted the work bench).

6.2 Place a metal sheet between the bottom platens and the belt

6.3 Align the splice once more with respect to the press platens.

6.4 Place the edge irons at both ends of the splice, which must be approximately 0.5 mm thinner than the belt.

6.5 Apply lateral pressure on the belt by holding the edge irons firmly against clamping bolts.

6.6 Place metal sheet and the top heating plate on the belt. Set down the top traverses on them and join firmly to the bottom traverses by clamping bolts.

6.7 Re-tighten the edge irons.

6.8 Switch off the heating platens and apply pressure gradually.

6.9 Apply full pressure (10-12 kg/cm²) when the platen temperature reaches 100°C.

6.10 Switch off the heating platen when the temperature reaches 140°C temp may go upto 143°C and will gradually came down. Switch it on again when it drops down to 140°C.

6.11 Tighten the press or check the hydraulic pressure continuously during the current period.

6.12 After full time cure, cool down the press to 60°C.

6.13 Dismantle the press, trim the spew, buff if necessary and then put the belt again in working condition.

7. Records

Keep the records of vulcanization in the enclosed Form no. 1.

N. B. 1) Plan and side view of prepared belt are shown in Sketch B.

2) Use jump type splice for two ply belts (Sketch C).

RECORDS OF VULCANISATION

Form No.1

Conveyor No.:

Date of Vulcanisation

Belt No.:

Splice Length:

Joint No.:

	TIME (MINS)	HYDRAULIC PRESSURE (KG/CM2)	TEMPERATURE (°C)			
			TOP PLATEN		BOTTOM PLATEN	
PREHEATING	0 min					
	5 min					
	10 min					

CURING						
COOLING						

Other Details:

Belt Construction

Remarks:

Name

Signature

Stamp:

Date

Splicing Team

Splicer

APPENDIX B

TECHNICAL INFORMATION FOR HOT SPLICING

1. Calculation of Splice Length

The extra length of belting required to form the splice can be calculated from the formula which follows:

$$\text{Additional Length (Ls)} = 0.3 W + \text{S.L.} (N-1) + 25\text{mm}$$

where, W = Belt width in mm

N = The number of carcass plies

and S.L = Step Length in mm

The step lengths of different belt constructions are given in Table 2. Splice steps are shown in Sketch A

2. INDUS range of splicing materials for splicing of textile conveyor belts by Hot Vulcanising method

The details are as follows:-

2.1 Cover Compound

Dimension	Quantity/Container	Use
1.6 mm X 175 mm X 4.5 Mtrs	1.5 Kgs	For all Grades

2.2 Insulation Compound

Dimension	Quantity/Container	Use
0.6 mm X 175 mm X 13 Mtrs	1.5 Kgs	For all Grades

2.3 Solution

Dimension	Quantity/Container	Use
	10 Ltrs.	For all Grades

2.4 Siliconised Paper

Dimension	Quantity/Container	Use
150 mm X 4mm	1 Roll	For all Grades

3. Specific Pressure required for Hot Splicing

10-12 Kg / cm² (or as required if hand-press is used)

4. Curing Temperature required for Hot Splicing

160±3°C.

5. Curing Time required for Hot Splicing

Belt Thickness (mm)	Time (min)	Cooling (°C)
Upto 10	20	Upto 80
Over 10 upto 15	30	upto 80
Over 15 upto 20	35	upto 80
Over 20 upto 25	40	upto 80
Over 25 upto 30	55	upto 80

6. Splicing Tools required

- 6.01 Knife with longer blade
- 6.02 Saddler's Knife
- 6.03 Cobbler's knife with retractable blade
- 6.04 Ply cutting knife
- 6.05 Knife with flexible blade
- 6.06 Angled knife
- 6.07 Quarter moon knife
- 6.08 Whet stone
- 6.09 Knurled roller
- 6.10 Flat roller
- 6.11 Wire brush
- 6.12 Scissors
- 6.13 Pincers
- 6.14 Screw Drivers
- 6.15 Hand brush
- 6.16 3 meter measuring tape
- 6.17 Trisquare

- 6.18 Rasp file
- 6.19 Surface Grinder
- 6.20 Soft - bristled brush
- 6.21 Steel ruler
- 6.22 White chalked string
- 6.23 Awl/ pricker
- 6.24 Iron mallet
- 6.25 Air blower

TABLE 2

Rubber Belt Splice Step Lengths for Hot Vulcanisation Method

FABRIC TYPE	RECOMMENDED STEP LENGTHS			
	2 PLY	3 PLY	4 PLY	5 PLY & MORE
PP 100 EP 100 28 Oz CN 70	150	150	--	---
PP 125 EP 125 EP 125 EP 160 32 Oz 34 Oz CN 80	150 200	200 200	200 200	200 200
PP 200 EP 200	200	250	250	250
PP 250 EP 250 36 OZ 42 OZ 48 OZ CN 90 CN 105 CN 130 CN 175 CN 210	---	300	300	300
PP 315 EP 315	---	350	350	350
PP350 EP 350 EP 400	---	400	400	400
EP 500	---	500	500	500

**CAUSES
&
CORRECTIONS**

SOME OF THE PROBLEMS ENCOUNTERED WITH CONVEYOR BELTING AND THE MEANS TO OVER COME THEM		
TROUBLES	CAUSES	CORRECTION
A. Excessive Edge wear, broken/frayed due to rubbing	1. Off-centre Loading 2. Material spillage and build up 3. Bowed Belt 4. Belt hitting structure 5. Misaligned running 6. Inadequate edge clearance 7. Belt too stiff for adequate troughing.	Adjust chute to place load on centre-of belt ; discharge material in direction of belt travel Improve loading and transfer conditions, install cleaning, devices, improve maintenance. For new belt, this condition should disappear during break- in; in rare instances belt must be straightened or replaced, check storage and handling of belt rolls. Install training idlers on carrying & returning run. Check conveyor alignment and re-align belt. Minimum recommended clearance between belt edge and structure is 75mm Install a more flexible belt.
B. Excessive top cover wear, uniform around belt width	1. Dirty, stuck or misaligned return roll 2. Cover quality too low 3. Material spillage and build-up. 4. Off-centre loading 5. Excessive sag between idlers and incorrect idler spacing.	Remove accumulations, install cleaning devices, use self-cleaning return rolls, Improve maintenance. Replace with belt of heavier cover, gauge or higher quality rubber. Same as A2 above Same as A1 above Idlers causing load to work and shuffle on belt as it passes over idlers; increase tension if unnecessarily low, reduce idle spacing.
C. Grooving, gauging or stripping of cover	1. Skirt boards improperly adjusted or of wrong doing materials. 2. Belt spanking down under load impact. 3. Material hanging under chute 4. Impact of materials on belt. 5. Worm or seized return idler roll	Adjust Skirt board support to minimum 25mm between metal and belt with gap increasing in direction of belt travel; use skirt board rubber (not old belt) Install cushion idlers. Improve loading to reduce spillage, install baffles; widen chute Reduce impact by improving chute designs, install impact idlers. Replace all unserviceables with new ones.

SOME OF THE PROBLEMS ENCOUNTERED WITH CONVEYOR BELTING AND THE MEANS TO OVER COME THEM		
TROUBLES	CAUSES	CORRECTION
D. Severe pulley cover wear	1. Sticking idlers 2. Slippage on drive pulley 3. Material spillage and build up 4. Material trapped between belt and pulley 5. Bolt heads protruding above lagging 6. Excessive forward tilt of trough rolls	Free idlers and improve maintenance Increase tension; lag driver pulling increase area of contact Same as A2 above Install scrapers on return run ahead of tail pulley Tighten bolts replace lagging use vulcaniser on lagging Reduces forward tilt of idlers to no more than 2° from vertical.
E. Cover harden or crack	1. Heat or chemical damage 2. Improper storage of handling	Use belt designed for specific condition Refer page 1 for proper storage and handling instructions.
F. Cover Softening/ Swells in spots or streaks	1. Contamination by spilled oil/ grease over lubrication of idlers	Improve housekeeping, reduce quantity of grease used, check grease seals.
1. Cover Blisters or sand blisters	1. Cover cuts or very small punctures allow fines to work under cover and cut cover away from carcass	Make spot repair with vulcaniser or self-curing repair material
2. Short breaks to carcass parallel to belt edge, star break in carcass.	1. Impact of material on belt	Same as C4 above
3. Transverse breaks	1. Belt edges folding up on structure. 2. Improper transition between troughed belt and terminal pulley 3. Severe Convex (Hump) vertical curve	Readjust idlers in affected area, install limit switches; provide more clearance Adjust transition as recommended Decrease idlers spacing in curve; increase curve radius
4. Lengthwise carcass breaks with cover intact.	1. Belt running off centre and folding over	Use limit switches and determine reasons for belt running out

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